👩🏾‍🏫 Staff Guide · Horticulture 1 · Unit 5 Soil & Nutrients (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 5 Environmental Requirements for Good Plant Growth (textbook Unit 4): media and soil, N-P-K nutrients (definitions, excess, deficiency), deficiency photo research, soil texture jar test, self-evaluation questions")	**Level 1 piece**
Textbook Unit 4 + Plant Media and Nutrition	Slides "Plants Need", "Growing Media" + notes A (facts rewritten)
Soil Analysis Parts 1-2 (jar test)	Soil Jar Test lab packet + notes B
Plant Nutrition Definitions (N-P-K, excess, deficiency)	Slides + notes C
Deficiency photo research	Slide "Deficiency Signs"
Unit 4 SE questions (16)	Modified quiz
Class slides	Deck: 16 slides, Words to Know first, "point to the picture" checks, recap
Her worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	9 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Jar test math: Level 1 says which layer is biggest; Level 2 uses a calculator frame
- Color anchors: 🟢 N, 🟣 P, 🟠 K

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part
- Cover one choice to make it 2

4. What to grade

- Soil analysis: Level 1 = photo + 'biggest layer'; Level 2 = layer % with calculator
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **Unit 4 SE question text**: The 16 self-evaluation questions are in the textbook scan only; our quiz is team-written.
- **Plant Media and Nutrition**: Title only in the export.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · N-P-K Plant Check (Level 1)

Horticulture 1 · Unit 5 · goes with the class “Deficiencies / Plant Nutrition GH Participation” (4 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Deficiencies / Plant Nutrition GH Participation”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Hand-over-hand or model first, then student tries
- Picture task card at the station
- Staff hold and pass every sharp tool

4. What to grade (4 pts)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: GH Participation nutrient activities were done in class and are not written in the export; this packet is team-written.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

👩🏾‍🏫 Staff Guide · Soil Jar Test (Level 1)

Horticulture 1 · Unit 5 · goes with the class “Soil Analysis Parts 1-2” (8 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Soil Analysis Parts 1-2”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do our hands go when the tool is cutting?
- What should we do when we finish?

3. Modifications

- Hand-over-hand or model first, then student tries
- Picture task card at the station
- Staff hold and pass every sharp tool

4. What to grade (8 pts)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
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- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
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- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

🪱 Soil & Nutrients Notes

Horticulture 1 · Unit 5 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

🪨 sand _____

🧱 clay _____

🌱 loam _____

🟢 nitrogen _____

🖼️ B. Fill in the word. Use the pictures.

5. 🪨 Big soil bits are _____.

6. 🧱 Tiny sticky bits are _____.

7. 🌱 The best mix is _____.

8. 🟢 N makes leaves _____.

Word bank: 🧱 **clay** | 🟢 **green** | 🌱 **loam** | 🪨 **sand**

⭕ C. Circle the right picture.

9. Plants need…



Soda



Water

10. Fertilizer in my mouth?



Yes



Never

11. In the jar, what sinks first?



Sand



Clay

🔗 D. Draw a line from the picture to the word.









silt

clay

loam

sand

📖 Soil & Nutrients: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Sand

big gritty soil bits



Silt

medium smooth bits



Clay

tiny sticky bits



Loam

the best mix for plants



Potting mix

soil for pots



Nitrogen (N)

makes leaves green



Phosphorus (P)

helps roots and flowers



Potassium (K)

keeps plants strong



Soil texture triangle

a chart for soil type

🟢 N-P-K Plant Check

Horticulture 1 · Unit 5 · Level 1

Name: _____

Modified from the class “Deficiencies / Plant Nutrition GH Participation”.

🧺 Supplies: get it. Check ☐.

- ☐  **Greenhouse plants** staff choose
- ☐  **Gloves**
- ☐  **N-P-K picture cards** yellow, purple, brown
- ☐  **Check chart**
- ☐  **Fertilizer scoop** staff only measures

🟢 Do one box. Check ☐ it.

1		Gloves on.	☐
2		Look at 3 plants.	☐
3	💛	Yellow old leaves? Point to N card.	☐
4	💜	Purple leaves? Point to P card.	☐
5	🤎	Brown edges? Point to K card.	☐
6	📋	Put a sticker on the chart.	☐
7		Wash hands.	☐

🚧 **Under construction:** GH Participation nutrient activities were done in class and are not written in the export; this packet is team-written. *Check with the class teacher.*

Soil Jar Test

Horticulture 1 · Unit 5 · Level 1



Name: _____

Modified from the class “Soil Analysis Parts 1-2”.

Supplies: get it. Check ☐.

- ☐ **Clear jar with lid**
- ☐ **Soil** about 2 inches
- ☐ **Water**
- ☐ **Ruler**
- ☐ **Gloves**
- ☐ **Calculator** Level 2

🫙 Do one box. Check ☐ it.

1		Gloves on.	☐
2		Put soil in the jar.	☐
3		Add water.	☐
4		Lid on tight.	☐
5		Shake, shake, shake!	☐
6		Wait until tomorrow.	☐
7		Point to the biggest layer.	☐

🟢 N-P-K Plant Check · Get Ready

Horticulture 1 · Unit 5 · Level 1

Name: _____

👥 My job. Circle it.

👩‍🌾 **Grower** plants	🧺 **Supply helper** gets tools	📸 **Recorder** photos	🧽 **Clean-up captain** cleans

🧰 Get the tools. Check ☐.

- ☐  Greenhouse plants
- ☐  Gloves
- ☐  N-P-K picture cards
- ☐  Check chart
- ☐  Fertilizer scoop

⚠️ Be safe!

- 🧤 Gloves on.
- 👄 Nothing in my mouth.
- ✂️ Tools with an adult.
- 🥄 Staff measure.
- 🧼 Wash hands.

✅ Before we start

☐ Gloves ☐ Apron ☐ Adult at station

🫙 Soil Jar Test · Get Ready

Horticulture 1 · Unit 5 · Level 1



Name: _____

👥 My job. Circle it.

👩‍🌾 **Grower** plants	🧺 **Supply helper** gets tools	📸 **Recorder** photos	🧽 **Clean-up captain** cleans

🧰 Get the tools. Check ☐.

- ☐ 🫙 Clear jar with lid
- ☐ 🪱 Soil
- ☐ 💧 Water
- ☐ 📏 Ruler
- ☐ 🧤 Gloves
- ☐ 🧮 Calculator

⚠️ Be safe!

- 🧤 Gloves on.
- 👄 Nothing in my mouth.
- ✂️ Tools with an adult.
- 😷 No dust in my face.
- 🧼 Wash hands.

✅ Before we start

☐ Gloves ☐ Apron ☐ Adult at station

🟢 N-P-K Plant Check · My Check

Horticulture 1 · Unit 5 · Level 1

Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🧤 I used gloves
- ☐ 🌱 I did the plant job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🧤 **I was safe.**	😀	🙂	😕
👩‍🌾 **I did my job.**	😀	🙂	😕
🌷 **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🫙 Soil Jar Test · My Check

Horticulture 1 · Unit 5 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🧤 I used gloves
- ☐ 🌱 I did the plant job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🧤 **I was safe.**	😀	🙂	😕
👩‍🌾 **I did my job.**	😀	🙂	😕
🌷 **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Soil & Nutrients Quiz

Horticulture 1 · Unit 5 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ☀️ Plants need…

- Darkness only
- Light

2. 🪨 The biggest soil bits are…

- Sand
- Clay

3. 🧱 The tiniest soil bits are…

- Clay
- Sand

4. 🌱 The best mix for plants is…

- All rocks
- Loam

5. 🟢 N helps…

- Music
- Green leaves

6. 🟣 P helps…

- 🌸 Roots and flowers
- 🚗 Cars

7. 🟠 K helps the plant be…

- 😴 Sleepy
- 💪 Strong

8. 🧤 With fertilizer I wear…

- 🧤 Gloves
- 🕶️ Sunglasses only

🪱 Soil & Nutrients Notes · KEY

Horticulture 1 · Unit 5 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

 sand _____

 clay _____

 loam _____

 nitrogen _____

🖼️ B. Fill in the word. Use the pictures.

5.  Big soil bits are **sand**.

6.  Tiny sticky bits are **clay**.

7.  The best mix is **loam**.

8.  N makes leaves **green**.

Word bank:  **clay** |  **green** |  **loam** |  **sand**

⭕ C. Circle the right picture.

9. Plants need…

 Soda	 Water

10. Fertilizer in my mouth?

 Yes	 Never

11. In the jar, what sinks first?

 Sand	 Clay

🔗 D. Draw a line from the picture to the word.

Key: 🪨 → sand · 🟤 → silt · 🧱 → clay · 🌱 → loam

	silt
	clay
	loam
	sand

🔑 *Key written by our team from the class teacher's materials, please check.*

✅ Soil & Nutrients Quiz · KEY

Horticulture 1 · Unit 5 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ☀️ Plants need… Unit 4 SE

- Darkness only
- ✅ Light

2. 🪨 The biggest soil bits are… Soil Analysis

- ✅ Sand
- Clay

3. 🧱 The tiniest soil bits are… Soil Analysis

- ✅ Clay
- Sand

4. 🌱 The best mix for plants is… Soil Analysis

- All rocks
- ✅ Loam

5. 🟢 N helps… Nutrition

- Music
- ✅ Green leaves

6. 🟣 P helps… Nutrition

- **Roots and flowers** 
- **Cars** 

7. 🟠 K helps the plant be… Nutrition

- **Sleepy** 
- **Strong** 

8. 🧤 With fertilizer I wear… Safety

- **Gloves** 
- **Sunglasses only** 

Answers

1-B (Light) · 2-A (Sand) · 3-A (Clay) · 4-B (Loam) · 5-B (Green leaves) · 6-A (Roots and flowers) · 7-B (Strong) · 8-A (Gloves)

Level 1 version of the gen ed “Plant Nutrition + Soil Analysis”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*